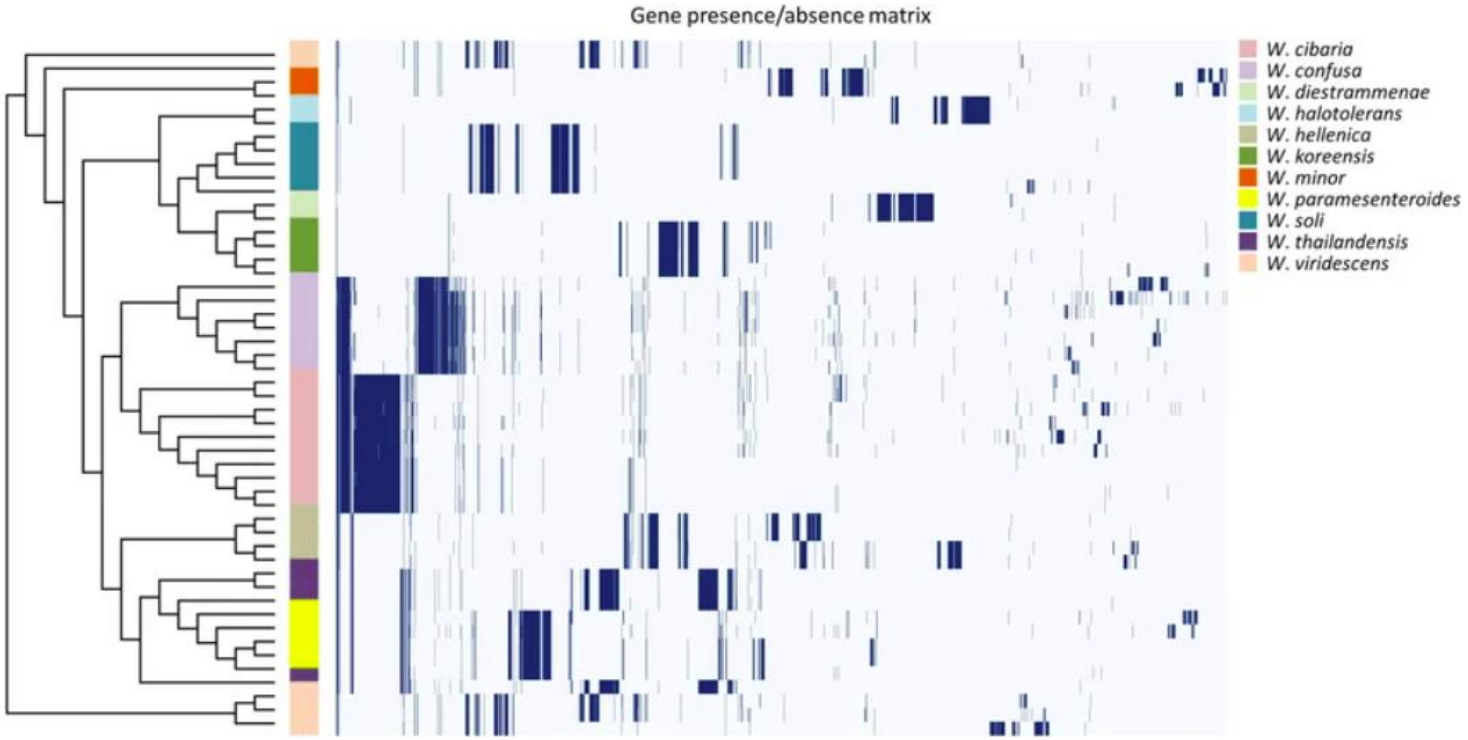
Day 8: Pangenomics

MMB-127

Pangenomics

A **pangenome** is the complete set of genes found across all strains of a species or group

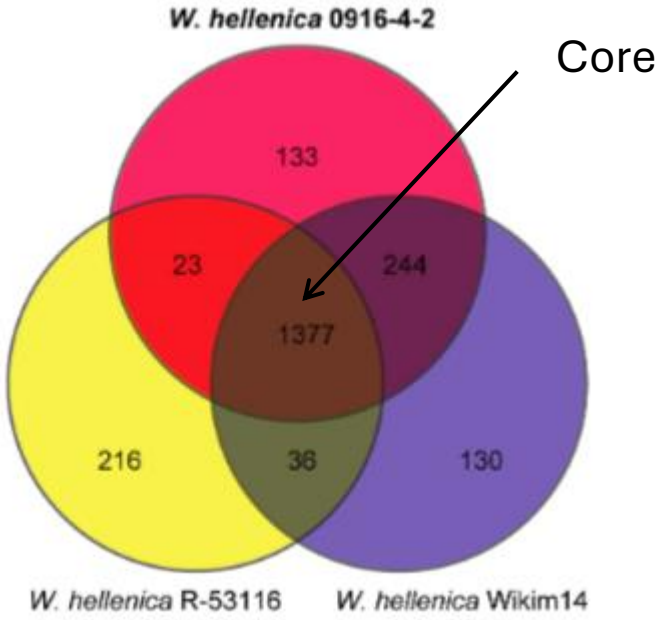


Pangenome analysis for 50 Weissella genomes using Roary. The left side represents the phylogenetic tree based on core-genome. The right side represents the matrix where the accessory and core genes were either present or absent

Kim et al., 2022

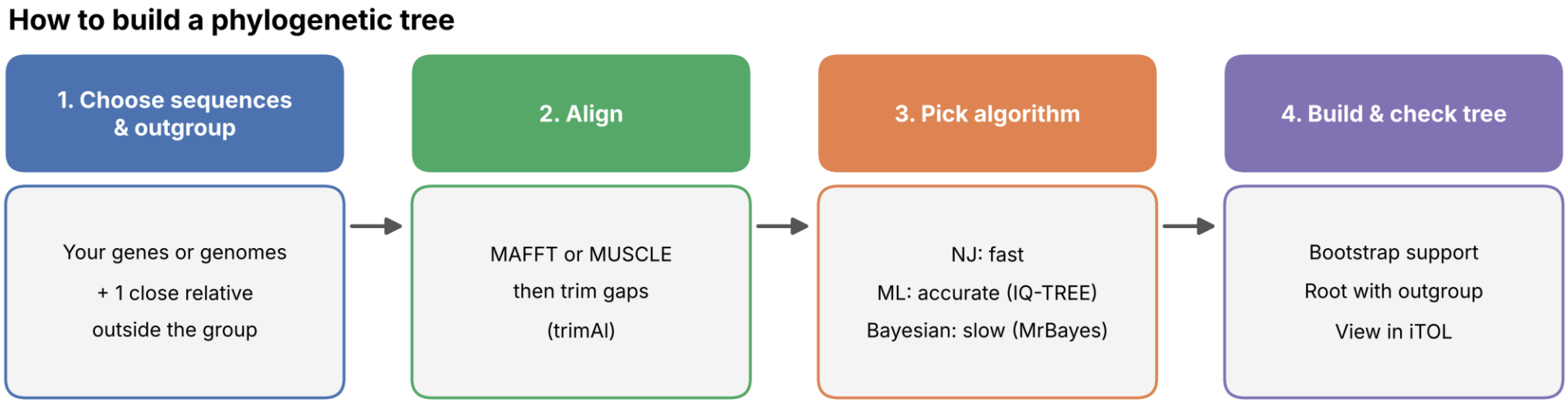
Core genes and accessory genes

- Core genome: genes present in all (or nearly all, e.g., ≥99%) genomes.
- Accessory genome: present in some genomes



Kim

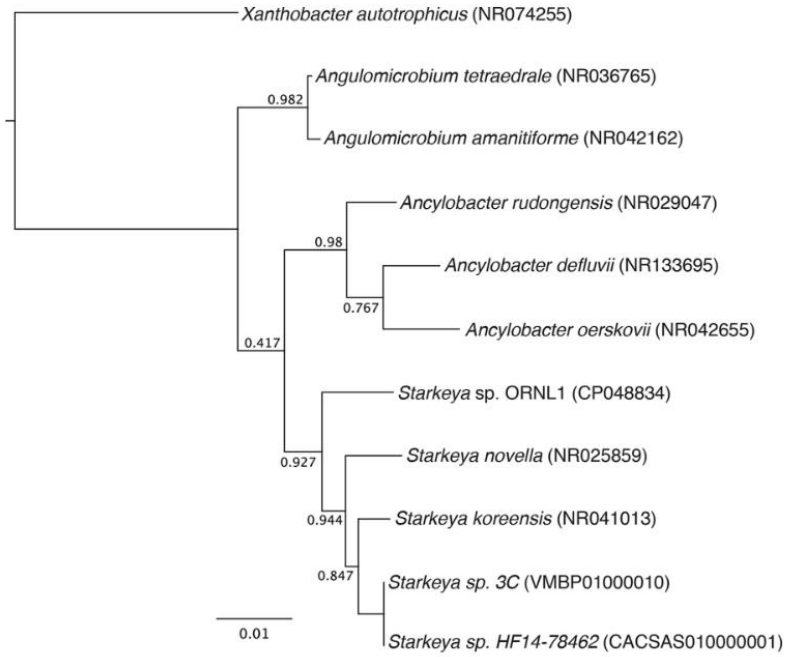
Building a phylogenetic tree



Building a phylogenetic tree

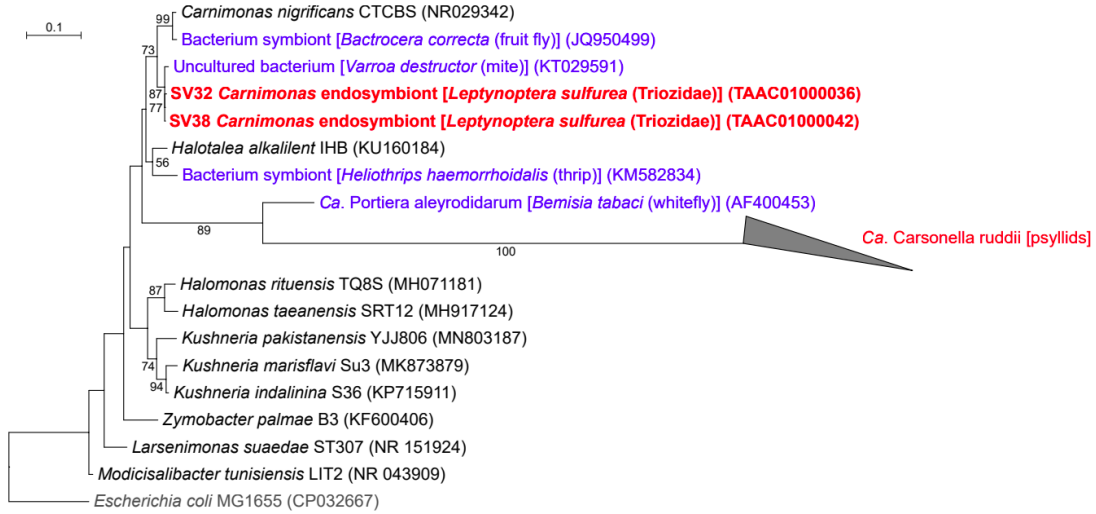
Bootstrap support tells you how strongly your data back each branch (clade)

FastTree calculates local support values using an SH-like (Shimodaira–Hasegawa-like) test (0-1).



Podar et al., 2020

IQ-TREE support values: Standard bootstrap (0-100).

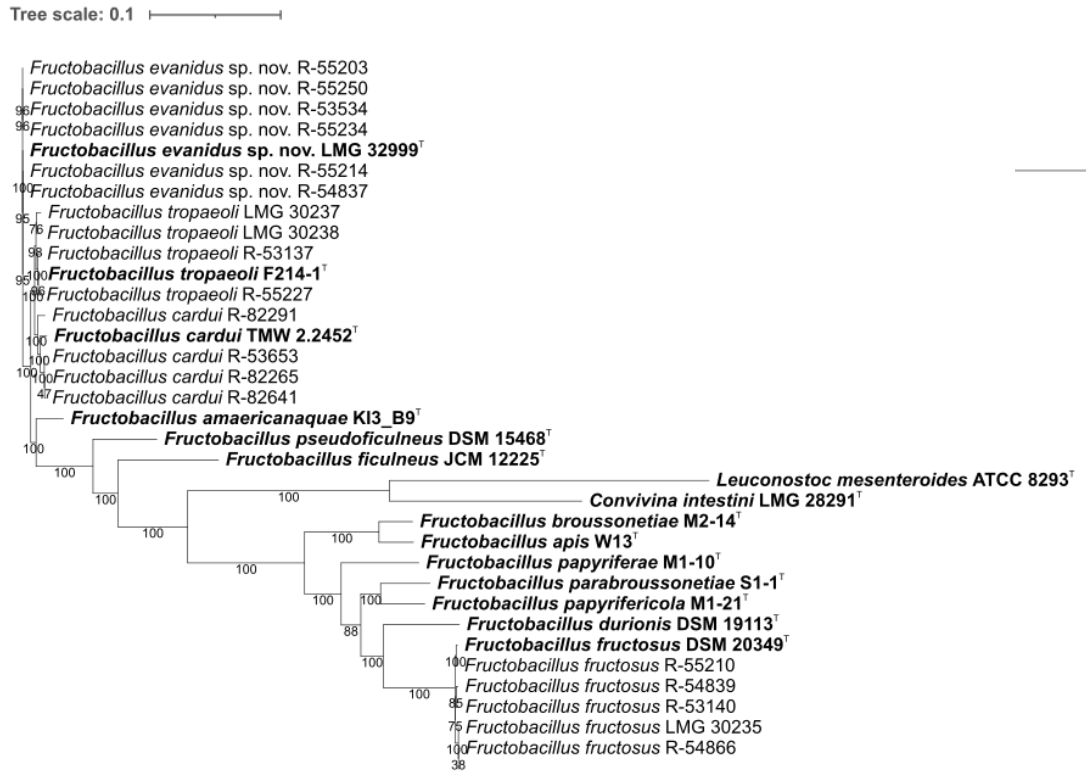


Nakabachi et al., 2022

Building a phylogenetic tree

- Root your phylogenetic tree

Unrooted tree



Rooted tree

